

Junjie Mao

jason080200114@gmail.com | [Portfolio](#)

Software Engineer with strong **foundations** in **Computer Science** and **Mathematics**. Experienced in building **full-stack** web applications with **C#** and **.NET** on the backend and **Vue.js/HTML/CSS** on the frontend, backed by **SQL** data management. Additional background in **AI code quality evaluation** and **GPU/graphics programming**.

PROFESSIONAL EXPERIENCE

[CinLearn Education](#) *Sept 2021 - Jun 2022*

Software Developer | Shanghai, China

- Developed backend services and **RESTful API** endpoints in **C#** and **.NET** for user authentication, course data management, and platform analytics.
- Built and maintained the platform's **frontend** using **HTML**, **CSS**, and **Vue.js**, delivering interactive course and student dashboards.
- Wrote and managed **SQL** queries to **store and retrieve** course and user data supporting platform features and analytics.
- Collaborated across **functional teams** to maintain and extend a **large**, production **codebase**.

[DataAnnotation](#) *Jun 2026 - Present*

Software Engineer Analyst – AI Trainer | Remote Freelance

- Agentically coded solution prompts using industry-standard **task formats** (i.e., Harbor), including feature creation, code migrations, and optimization from open-source libraries for **AI training**.
- Evaluated and analyzed complex **C++**, **Python** and **Go** code solutions, and the **Trajectories** generated by multiple **frontier language models** (i.e. gemini, claude, chat gpt).

[Raylib](#) *2026*

Game Programmer, Contributor

- Worked inside a general-purpose **C Game library (34.7k stars on GitHub)** similar to [SDL](#) and [XNA](#).
- **Diagnosed out-of-bounds reads** in the **rcore.c** input module in `IsMouseButtonPressed/Down/Released/Up()` that indexed the mouse button state arrays without validating the argument, **causing the out-of-range read** that leads to nondeterministic behaviour.
- **Isolated the bug** with a **minimal reproduction program** and **screen capture**, then **submitted a fix** that was **merged** by the project maintainer in [PR #5516](#).
- **Reworked the patch** in review to match **project conventions** (no **early returns**, consistent with the `IsKey*()` family), and **identified an error** in the maintainer's suggested implementation before it landed.

[RayGui](#) *2025*

Game Tool Programmer, Contributor

- **Worked** inside a single-header **C library (~5k LOC, 5.1k stars on GitHub)** that consists of an immediate-mode UI controls backend.
- **Debugged** an **MSVC C4701 compiler warning** in `GuiTextBox` and submitted a **fix** that was **merged** by the project maintainer in [PR #515](#).

PROJECTS & TECHNICAL EXPERTISE

Vulkan GPU Renderer Engine | C++, Vulkan, GLFW, CMake, ImGui

[\[GitHub\]](#)

- Developed a **Vulkan-based 3D renderer in C++**, implementing swapchain management, render passes, graphics pipelines, descriptor sets, command buffers, synchronization, depth buffering, and MSAA.
- Built a GPU resource pipeline for **OBJ models and textures**, using **staging buffers** for device-local uploads, with explicit buffer/image transfers, pipeline barriers, image-layout transitions, and runtime GPU mipmap generation.
- Integrated **Dear ImGui as an in-engine rendering/debug interface**, providing a dockable scene viewport and runtime controls for camera, fog, rendering parameters, and model counts.
- Added **shader hot-reloading** that detects source changes and recreates the graphics pipeline, allowing shader changes to be tested without restarting the renderer.

Jason's 3rd Law | C++, Raylib, Windows API

[\[GitHub\]](#)

- Developed a **2D puzzle game in C++** as the **sole programmer on a two-person team**, implementing gameplay systems, entity behavior, level loading, rendering, input, audio, UI, and game-state management.
- Designed a **contiguous entity storage and indexed lookup system** using a central entity array and per-layer lookup tables to efficiently organize and access entities across puzzle systems.
- Built custom development infrastructure including **persistent/transient bump allocators, fixed-capacity arrays, and Windows DLL hot-reloading**, enabling gameplay code to be recompiled and updated while the game remained running
- Built a **Tiled level-loading pipeline** that parses **JSON** files that consist of **tilemaps and tile properties** to construct entities, configure level metadata, initialize gameplay state, and organize entities into runtime layers.
- Implemented **undo/revert gameplay state management**, maintaining previous game states and restoring entity/gameplay state when the player reverses an action.
- Implemented complex puzzle interactions involving **movable entities, mass-dependent movement, directional constraints, cables and electrical connections, doors, portals, switches, locks, and environmental hazards**.
- Developed the game's **movement and collision logic**, including entity movement, grid-based positioning, and state-dependent interactions between entities.

Portal Demo | Unity, C#, Character Controller, RenderTexture

[\[GitHub\]](#), [\[Demo\]](#)

- Implemented a portal **rendering system** in Unity using **transform-space matrix mapping** (localToWorldMatrix / worldToLocalMatrix) to position portal cameras, dynamically create screen-sized **RenderTextures**, and feed the resulting views into **portal materials and shaders**.
- Implemented **portal traversal** for players and **grabbable objects** using **Unity Physics** for collision checks and receiver-relative position transforms.
- Built a custom **Rigidbody-based first-person controller** with **Unity Input System, acceleration and friction-based movement**.

SKILLS

Languages: C#, JavaScript, SQL, Python, C++, Go, GDScript, R, GPU Shaders (GLSL/HLSL)

Web & Frontend: .NET, Vue.js, HTML, CSS

Backend & Systems: Compiler Design, Multithreading, REST APIs, GPU Rendering Pipelines, OpenGL, Vulkan

AI: AI Code Evaluation, Harbor task format, Data Science

DevOps & Tools: Git, Linux/Unix, Bash, MS-DOS, Docker, CI/CD Pipelines, GitHub Actions, Unity, Unreal, Godot

EDUCATION

University of Waterloo, Waterloo, Ontario

Bachelor of Math, Honours Computer Science, and Joint Honours Combinatorics and Optimization

September 2020 - September 2024

Coursework: Compiler Construction, Data Structures & Algorithms, Statistics, Operating Systems, Distributed Systems.